A counterexample to Stanley's conjecture on Smith forms of Jucys–Murphy elements

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Abstract

For a partition λ of n with f^λ standard Young tableaux, let $S^\lambda_{\mathbb{Z}}$ be the integral Specht module and $X_k = (1,k) + \cdots + (k-1,k)$ the k-th Jucys–Murphy element. Stanley conjectured in 2016 that the invariant factors of X_k on $S^\lambda_{\mathbb{Z}}$, taken nonnegative, are $\alpha_{f^\lambda - r + 1} = |\prod(j-i)|$ for $1 \le r \le f^\lambda$, the product (empty product 1) over the cells (i,j) that k occupies in at least r standard tableaux. We show that it is false. In $\mathbb{Z}\mathfrak{S}_n$, $X_3\big(X_3 - (1,2)\big) = 2$. Hence, for $n \ge 3$, on every full-rank $\mathbb{Z}$-lattice in $S^\lambda_{\mathbb{Q}}$ stable under the permutations of $\{1,2,3\}$, including $S^\lambda_{\mathbb{Z}}$, X_3 has Smith form $(1^{f^\lambda - m}, 2^m)$, where m counts the tableaux with 3 in cell $(1,3)$ or $(3,1)$. So the conjecture holds at $k = 3$ exactly when $\lambda_1 \le 2$ or λ has at most two rows; the smallest failure, $\lambda = (3,1,1)$, gives $(1,1,1,1,2,2)$ against the predicted $(1,1,1,1,1,4)$, and no case with $n \le 4$ fails. Lean checks the identity and the 6×6 matrix computation for $(3,1,1)$, independent programs check that this matrix represents X_3, and the general argument is a written proof.

The proofs, the Lean formalization, the certificate checkers and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

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1 Introduction

In his survey *Smith normal form in combinatorics* [8], Richard P. Stanley proposed studying the Smith normal form (SNF) of natural elements of the group ring $\mathbb{Z}\mathfrak{S}_n$. His example is the Jucys–Murphy element $X_k = (1,k) + (2,k) + \cdots + (k-1,k)$ [4, 6], acting on the Specht module S^λ, where its eigenvalues are the contents of the cells occupied by k in the standard Young tableaux of shape λ. The survey reports that Darij Grinberg computed the SNF of the matrices $W_{\lambda,k}$ for $\lambda \vdash n \le 7$, and on the basis of these data Stanley formulated Conjecture 7.4: the Smith form is given by a product formula in the contents, recalled exactly in Section 2. Stanley illustrated it with $\lambda = (5,1)$, $k = 5$, where the formula gives $(1,1,3,3,12)$.

On December 25, 2024, Alexander Chervov asked on MathOverflow about the status of these conjectures [1]; the question had no answer as of October 21, 2026. In a comment on it, Grinberg noted that the survey defines the matrices $W_{\lambda,k}$ "somewhat clumsily", through a basis of $\mathbb{Z}\mathfrak{S}_n$, and clarified the intended lattice: the integral Specht module with its standard polytabloid basis. We use exactly this reading.

Our main observation is an identity in $\mathbb{Z}\mathfrak{S}_3 \subseteq \mathbb{Z}\mathfrak{S}_n$ (Lemma 3):

$$X_3(X_3 - (1,2)) = (X_3 - (1,2))X_3 = 2.$$

It forces every invariant factor of X_3 to divide 2, on every lattice stable under the permutations of $\{1,2,3\}$.

Theorem 1. *Let $\lambda \vdash n \geq 3$, and let m be the number of standard Young tableaux of shape λ in which 3 occupies the cell $(1,3)$ or $(3,1)$. On every full-rank $\mathbb{Z}$-lattice in $S^\lambda_\mathbb{Q}$ that is stable under the permutations of $\{1,2,3\}$, in particular on $S^\lambda_\mathbb{Z}$, the Smith normal form of X_3 is $(1^{f^\lambda - m}, 2^m)$.*

Corollary 2. *Conjecture 7.4 holds at $k=3$ if and only if $\lambda_1 \leq 2$ or λ has at most two rows. When $\lambda_1 \geq 3$ and λ has at least three rows, it fails on every lattice as in Theorem 1: the predicted largest invariant factor is 4, while every actual one divides 2.*

The smallest such shape is $\lambda = (3,1,1)$. There the Smith form of X_3 on $S^{(3,1,1)}_\mathbb{Z}$ is $(1,1,1,1,2,2)$, while Conjecture 7.4 predicts $(1,1,1,1,1,4)$; Section 4 gives the 6×6 matrix explicitly. No case with $n \leq 4$ fails, and at $n = 5$ the only failures are $\lambda = (3,1,1)$ with $k = 3$ and with $k = 5$ (Proposition 5); these are the smallest counterexamples in this sense. A census of all $\lambda \vdash n \leq 8$ and all k finds 94 disagreements among 416 cases (Section 8).

What is checked, and how. Lean proves the identity of Lemma 3 in the group ring and, for the explicit 6×6 integer matrix A of Section 4, its Smith form, the relation $AB = 2I$ with an explicit integer matrix B, and that the predicted form is not reachable from A. That A is the matrix of X_3 on $S^{(3,1,1)}_\mathbb{Z}$ in the standard polytabloid basis is established by two independent certificate checkers, each working from the definitions of tableaux, tabloids and polytabloids rather than trusting the encoded matrix: one recomputes A, the other verifies the defining equations $X_3 e_{T_j} = \sum_i A_{ij} e_{T_i}$ in the tabloid module, which determine A. Theorem 1, Corollary 2 and the hand argument for $n \leq 4$ are written proofs and are not machine-checked; a third checker recomputes every case with $n \leq 5$. Checker A also verifies, for this λ, that the standard polytabloids form a $\mathbb{Z}$-basis of the span of all polytabloids, so the witness does not rest on the standard basis theorem; the two checkers enumerate the standard tableaux by different methods. The census for $n \leq 8$ (Sections 6–8) is supporting data, not part of any proof: it is computed by a construction that uses no standard basis and agrees with a second, standard-basis computation for all $n \leq 6$; its programs and output are included. Section 10 gives details.

Credits. The conjecture, its formulation and its worked example are Stanley's. The computations on which it rests are reported in the survey as Grinberg's, and the clarification of the intended lattice is his. Chervov's question drew attention to the status of the conjecture. We know of no earlier report that the conjecture fails. We do not know why the computations reported in the survey supported the conjecture, and we attribute no error to them (Section 9).

2 The statement and the reading used

The following is quoted from the final part of Section 7 of [8] (arXiv version 2, page 15; the text of version 1 is the same). Reference numbers inside quotations are those of the quoted source.

> One intriguing example is the *Jucys-Murphy element* (though it first appears in the work of Alfred Young [50, §19]) $X_k \in \mathbb{Q}\mathfrak{S}_n$, $1 \leq k \leq n$. It is defined by $X_1 = 0$ and
>
> $$X_k = (1,k) + (2,k) + \cdots + (k-1,k), \quad 2 \leq k \leq n,$$

where (i,k) denotes the transposition interchanging i and k. Just as for $\Gamma_n(q)$, we can choose an integral basis for $\mathbb{Q}\mathfrak{S}_n$ (that is, a $\mathbb{Z}$-basis for $\mathbb{Z}\mathfrak{S}_n$) so that the action of X_k on $\mathbb{Q}\mathfrak{S}_n$ with respect to this basis has a matrix of the form $\bigoplus_{\lambda\vdash n} f^\lambda W_{\lambda,k}$. The eigenvalues of $W_{\lambda,k}$ are known to be the contents of the positions occupied by k in all standard Young tableaux of shape λ. [...] Darij Grinberg (private communication) computed the SNF of the matrices $W_{\lambda,k}$ for $\lambda \vdash n \le 7$. On the basis of this data we make the following conjecture.

CONJECTURE 7.4. *Let $\lambda \vdash n$, $1 \le k \le n$, and $W_{\lambda,k} \overset{\text{snf}}{\to} (\alpha_1, \dots, \alpha_{f^\lambda})$. Fix $1 \le r \le f^\lambda$. Let S_r be the set of positions (i,j) that k occupies in at least r of the SYT's of shape λ. Then $\alpha_{f^\lambda - r + 1} = \pm \prod_{(i,j)\in S_r} (j-i)$.*

Note in particular that every SNF diagonal entry is (conjecturally) a product of some of the eigenvalues of $W_{\lambda,k}$.

For example, when $\lambda = (5,1)$ and $k = 5$ we have $f^{(5,1)} = 5$ and $S_1 = \{(1,5),(2,1),(1,4)\}$, $S_2 = S_3 = \{(1,4)\}$, $S_4 = S_5 = \emptyset$. Hence $W_{(5,1),5} \overset{\text{snf}}{\to} (1,1,3,3,12)$.

The omitted passage lists the standard Young tableaux of shape $(5,1)$ and their eigenvalues. Here f^λ is the number of standard Young tableaux (SYT) of shape λ, the cell (i,j) is in row i and column j, and $j-i$ is its content. The invariant factors $\alpha_1 \mid \alpha_2 \mid \cdots$ are taken nonnegative, and an empty product is 1.

In his comment of December 25, 2024 on [1], Grinberg wrote (the emphasis, bold in the original, is set in italics here):

> Note about Question 1: The $W_{\lambda,k}$ in §7 of Stanley's paper is defined somewhat clumsily. It should be defined as the matrix representing the action of the YJM element X_k on the Specht module $\mathcal{S}^\lambda$ over $\mathbb{Z}$ with respect to the standard tableau basis. You can, of course, take any other basis, since the Smith normal form is conjugation-invariant. But the direct sum $\bigoplus_{\lambda\vdash n} f^\lambda W^{\lambda,k}$ is *not* the action of X_k on the group ring of S_n, so you cannot define it through this sum.

Question 1 of [1], to which the note refers, reads: "Is there any progress on these conjectures ?" In a second comment Grinberg added that he was "still interested in Question 1."

The reading used. We follow Grinberg's clarification. Let $M^\lambda_{\mathbb{Z}}$ be the free $\mathbb{Z}$-module on the row tabloids $\{T\}$ of shape λ, with $\mathfrak{S}_n$ acting by $\pi\{T\} = \{\pi T\}$, where πT replaces each entry x of T by $\pi(x)$. For a tableau T with column stabilizer C_T, the polytabloid is $e_T = \sum_{\pi\in C_T} \operatorname{sgn}(\pi)\,\pi\{T\}$, and the integral Specht module $S^\lambda_{\mathbb{Z}}$ is the $\mathbb{Z}$-span of all e_T. The standard polytabloids e_T, T standard, form a $\mathbb{Z}$-basis of $S^\lambda_{\mathbb{Z}}$ ([3, §8], [7, §2.6]); for the cases we compute, this is also verified directly by the checkers (Section 10). Then $W_{\lambda,k}$ is the matrix of X_k in the standard polytabloid basis, and its Smith form is that of X_k as an endomorphism of $S^\lambda_{\mathbb{Z}}$, which does not depend on the basis. By a lattice in $S^\lambda_{\mathbb{Q}}$ we mean a full-rank $\mathbb{Z}$-lattice, that is, a $\mathbb{Z}$-submodule of rank f^λ spanning $S^\lambda_{\mathbb{Q}}$; the Smith form of an endomorphism of a lattice is that of its matrix in any $\mathbb{Z}$-basis. Section 6 records what happens under other readings.

3 The Jucys–Murphy element X_3

Lemma 3. *In $\mathbb{Z}\mathfrak{S}_n$, $n \ge 3$, let $u = (1,2)$. Then $X_3(X_3 - u) = (X_3 - u)X_3 = 2$.*

Proof. Write $s = (1,3)$ and $t = (2,3)$, so $X_3 = s + t$. Let c and c^2 be the two 3-cycles on $\{1,2,3\}$. Then $X_3^2 = s^2 + t^2 + st + ts = 2 + c + c^2$, since st and $ts = (st)^{-1}$ are the two 3-cycles. Each of su, tu, us, ut is a product of two distinct transpositions in $\mathfrak{S}_3$, hence a 3-cycle, and $su \neq tu$, $us \neq ut$. So $X_3 u = u X_3 = c + c^2$, and $X_3(X_3 - u) = (X_3 - u)X_3 = 2$. □

Proof of Theorem 1. Let L be a lattice in $S^\lambda_{\mathbb{Q}}$ stable under the permutations of $\{1,2,3\}$. Both X_3 and $X_3 - u$ lie in $\mathbb{Z}\mathfrak{S}_3$, so in a $\mathbb{Z}$-basis of L they act by integer matrices A and B, and $AB = BA = 2I$ by Lemma 3. Choose unimodular P, Q with $PAQ = D = \operatorname{diag}(d_1, \ldots, d_f)$, $f = f^\lambda$. Then $D \cdot (Q^{-1}BP^{-1}) = 2I$ with $Q^{-1}BP^{-1}$ an integer matrix, so $d_i C_{ii} = 2$ for an integer C_{ii}, and each d_i is 1 or 2. The number of d_i equal to 2 is the 2-adic valuation of $|\det A|$.

The determinant of X_3 on $S^\lambda_{\mathbb{Q}}$ does not depend on L. By the classical theorem of Jucys and Murphy [4, 6], X_3 acts diagonally on Young's seminormal basis of $S^\lambda_{\mathbb{Q}}$, with eigenvalue on the vector indexed by a standard tableau T equal to the content of the cell of 3 in T. In a standard tableau the entries $1, 2, 3$ fill a partition of 3, so 3 occupies $(1,2)$, $(2,1)$, $(1,3)$ or $(3,1)$, of contents $1, -1, 2, -2$. Hence $|\det A| = 2^m$, and the Smith form is $(1^{f-m}, 2^m)$. □

Proof of Corollary 2. For a cell c let N_c be the number of standard tableaux of shape λ in which 3 occupies c ($N_c = 0$ if $c \notin \lambda$), so that $m = N_{(1,3)} + N_{(3,1)}$. Every S_r is contained in $\{(1,2), (2,1), (1,3), (3,1)\}$, and the first two cells have contents ± 1. Hence the prediction is

$$\alpha_{f-r+1} = 2^{[N_{(1,3)} \geq r] + [N_{(3,1)} \geq r]} \qquad (1 \leq r \leq f),$$

where $[\,\cdot\,]$ is 1 if the condition holds and 0 otherwise.

Suppose $\lambda_1 \geq 3$ and λ has at least three rows. The standard fillings of size three that place $1, 2, 3$ along the first row, or down the first column, put 3 in $(1,3)$, respectively $(3,1)$; both shapes lie in λ, and each filling extends to a standard tableau of shape λ by filling the remaining cells with $4, \ldots, n$ row by row, top to bottom and left to right. So both contents 2 and -2 occur, $N_{(1,3)} \geq 1$ and $N_{(3,1)} \geq 1$, and the prediction gives $\alpha_f = 4$. By Theorem 1 every invariant factor divides 2, so the conjecture fails, on every lattice of the theorem.

Otherwise at most one of $(1,3)$ and $(3,1)$ lies in λ, and the corresponding N_c equals m (if neither does, $m = 0$). The prediction is then $\alpha_{f-r+1} = 2$ for $r \leq m$ and 1 for $r > m$, that is, $(1^{f-m}, 2^m)$, which is the Smith form given by Theorem 1. □

Theorem 1 and Corollary 2 are written proofs and are not machine-checked; the identity of Lemma 3 is checked in Lean. The census of Section 8 agrees with both for $n \leq 8$: in all 63 cases with $k = 3$ every invariant factor is 1 or 2, there are exactly m factors equal to 2, and the disagreements are exactly the shapes with $\lambda_1 \geq 3$ and at least three rows.

Remark 4. Over $\mathbb{Z}_{(2)}$, the idempotent $P = \frac{1}{3}(1 + c + c^2)$ gives the same conclusion in another form: $X_3^2 = 1 + 3P$, so X_3 is invertible on $(1 - P)L_{(2)}$ and acts as $2(1,3)$ on $PL_{(2)}$. Comparing the two descriptions gives $m = (f^\lambda + 2\chi^\lambda((1,2,3)))/3$, where χ^λ is the character of $S^\lambda_{\mathbb{Q}}$.

4 The smallest counterexample

Let $\lambda = (3,1,1)$, so $n = 5$, $f^\lambda = 6$ and $X_3 = (1,3) + (2,3)$. Order the standard tableaux as

$$T_1 = \begin{array}{|c|c|c|}\hline 1&2&3\\\hline 4\\\cline{1-1} 5\\\cline{1-1}\end{array}\,, \quad T_2 = \begin{array}{|c|c|c|}\hline 1&2&4\\\hline 3\\\cline{1-1} 5\\\cline{1-1}\end{array}\,, \quad T_3 = \begin{array}{|c|c|c|}\hline 1&2&5\\\hline 3\\\cline{1-1} 4\\\cline{1-1}\end{array}\,, \quad T_4 = \begin{array}{|c|c|c|}\hline 1&3&4\\\hline 2\\\cline{1-1} 5\\\cline{1-1}\end{array}\,, \quad T_5 = \begin{array}{|c|c|c|}\hline 1&3&5\\\hline 2\\\cline{1-1} 4\\\cline{1-1}\end{array}\,, \quad T_6 = \begin{array}{|c|c|c|}\hline 1&4&5\\\hline 2\\\cline{1-1} 3\\\cline{1-1}\end{array}\,,$$

and let column j of A hold the coordinates of $X_3 e_{T_j}$ in the standard polytabloid basis, so that $X_3 e_{T_j} = \sum_i A_{ij} e_{T_i}$. Let $B = 2A^{-1}$; by Lemma 3 it is the matrix of $X_3 - (1,2)$ in the same basis, and it is integral. Then

$$A = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & -2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 2 & 0 & 0 \\ -1 & 0 & 2 & 0 & 2 & 0 \\ 0 & -1 & -1 & -1 & -1 & -1 \end{pmatrix},$$

and $AB = BA = 2I$. For example, the first column of A says $X_3 e_{T_1} = 2e_{T_1} - e_{T_2} + e_{T_3}$.

The prediction. The entry 3 occupies the cell $(1,3)$ in T_1, $(2,1)$ in T_2 and T_3, $(1,2)$ in T_4 and T_5, and $(3,1)$ in T_6. Hence $S_1 = \{(1,2),(1,3),(2,1),(3,1)\}$ with $|\prod(j-i)| = |1 \cdot 2 \cdot (-1) \cdot (-2)| = 4$, $S_2 = \{(1,2),(2,1)\}$ with product of absolute value 1, and $S_r = \emptyset$ for $3 \le r \le 6$. Conjecture 7.4 therefore predicts $(1,1,1,1,1,4)$.

The Smith form. By Theorem 1 with $m = 2$ (T_1 and T_6), the Smith form is $(1,1,1,1,2,2)$. The explicit matrices confirm this. With

$$U = \begin{pmatrix} 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ -1 & 0 & 2 & 0 & 2 & 0 \\ 0 & -1 & -1 & -1 & -1 & -1 \end{pmatrix}, \qquad V = \begin{pmatrix} 1 & -1 & 1 & 1 & -1 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

one has $\det U = -1$, $\det V = 1$ and $UAV = \mathrm{diag}(1,1,1,1,2,2)$. Without transformation matrices: A is lower triangular, so $\det A = 2 \cdot (-1) \cdot (-1) \cdot 1 \cdot 1 \cdot (-2) = -4$. Modulo 2, the first row of A vanishes and the sum of rows 2 to 6 vanishes, while rows 2 to 5 restricted to columns 2 to 5 form a unitriangular matrix; hence A has rank 4 over $\mathbb{F}_2$. So exactly two invariant factors are even, their product with the others is 4, and the form is $(1,1,1,1,2,2)$. The predicted form has only one even factor and would require rank 5 over $\mathbb{F}_2$.

By Theorem 1, the same Smith form arises on any lattice stable under the permutations of $\{1,2,3\}$ in a module isomorphic to $S^{(3,1,1)}_{\mathbb{Q}}$: for instance the lattices given by polytabloids built from column tabloids, by a left ideal generated by a Young symmetrizer, by the dual lattice, or by the saturated lattice $S^{\lambda}_{\mathbb{Q}} \cap M^{\lambda}_{\mathbb{Z}}$. No choice of lattice or basis of this kind restores the prediction.

A derivation by hand. The matrix can also be obtained without coordinates in the tabloid module. Let $e_1, \dots, e_5$ be the standard basis of $\mathbb{Z}^5$, with $\mathfrak{S}_5$ permuting it, and let $A_4 = \{x \in \mathbb{Z}^5 : \sum_i x_i = 0\}$. The map φ sending the tabloid with x in row 2 and y in row 3 to $e_x \wedge e_y \in \Lambda^2 \mathbb{Z}^5$ is $\mathfrak{S}_5$-equivariant. If T has first column (c_1, c_2, c_3), then C_T is the symmetric group on $\{c_1, c_2, c_3\}$, and summing the six signed terms gives

$$\varphi(e_T) = 2\,(e_{c_2} - e_{c_1}) \wedge (e_{c_3} - e_{c_1}) \in 2\Lambda^2 A_4.$$

For standard T the first column is $(1, a, b)$ with $2 \le a < b \le 5$, each pair occurring once, so the six standard polytabloids are sent to $2 f_a \wedge f_b$, where $f_i = e_i - e_1$. These form twice a $\mathbb{Z}$-basis of

$\Lambda^2 A_4$. Since $S^\lambda_{\mathbb{Q}}$ is irreducible and $\varphi \neq 0$, the map is injective on it, and $\frac{1}{2}\varphi$ is an $\mathfrak{S}_5$-equivariant isomorphism $S^{(3,1,1)}_{\mathbb{Z}} \cong \Lambda^2 A_4$. Now use the basis $b_i = e_i - e_5$ $(1 \le i \le 4)$ of A_4 and write $b_{ij} = b_i \wedge b_j$. The transpositions $(1,3)$ and $(2,3)$ fix b_4 and permute b_1, b_2, b_3, so X_3 preserves the span of b_{12}, b_{13}, b_{23} and the span of b_{14}, b_{24}, b_{34}, with matrices

$$\begin{pmatrix} 0 & 1 & -1 \\ 1 & -1 & 0 \\ -1 & 0 & -1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Their determinants are 2 and -2, and each has a 2×2 minor equal to ± 1, so each has Smith form $(1,1,2)$. Together they give $(1,1,1,1,2,2)$ again.

5 The cases with $n \le 5$

Proposition 5. *Conjecture 7.4 holds for every $\lambda \vdash n \le 4$ and every $1 \le k \le n$. For $n = 5$ it fails exactly for $\lambda = (3,1,1)$ with $k = 3$ and with $k = 5$.*

For $\lambda = (3,1,1)$ and $k = 5$ the Smith form is $(1,1,2,2,4,4)$ and the prediction $(1,1,1,4,4,4)$. The first statement has a written proof, given below, which works on every lattice in $S^\lambda_{\mathbb{Q}}$ preserved by X_k. Both statements are also verified by a standalone checker that recomputes all 69 cases with $n \le 5$ (Section 10), and by the census.

Proof for $n \le 4$. We use again that X_k acts diagonally on Young's seminormal basis, with eigenvalues the contents of the cells occupied by k in the standard tableaux [4, 6]. The determinant and the rational rank of $X_k - a$ $(a \in \mathbb{Z})$ therefore do not depend on the lattice.

If $k = 1$, then $X_1 = 0$, and 1 occupies $(1,1)$, of content 0, in every tableau; the prediction is $(0, \dots, 0)$. If $k = 2$, then $X_2 = (1,2)$ is an involution, hence unimodular on every lattice, and its Smith form is $(1, \dots, 1)$; the entry 2 occupies $(1,2)$ or $(2,1)$, of content ± 1, so every predicted factor is 1. If $f^\lambda = 1$, then $\lambda = (n)$ or (1^n), X_k is the scalar $\pm(k-1)$, and k occupies $(1,k)$ or $(k,1)$, as predicted. The remaining cases are listed below, with the cells occupied by k in the standard tableaux (multiplicities as exponents) and the prediction.

λ, k	cells of k	eigenvalues	prediction
$(2,1)$, 3	$(1,2)$, $(2,1)$	$1, -1$	$(1,1)$
$(2,2)$, 3	$(1,2)$, $(2,1)$	$1, -1$	$(1,1)$
$(2,2)$, 4	$(2,2)^2$	$0, 0$	$(0,0)$
$(3,1)$, 3	$(1,3)$, $(1,2)$, $(2,1)$	$2, 1, -1$	$(1,1,2)$
$(2,1,1)$, 3	$(3,1)$, $(1,2)$, $(2,1)$	$-2, 1, -1$	$(1,1,2)$
$(3,1)$, 4	$(1,3)^2$, $(2,1)$	$2, 2, -1$	$(1,2,2)$
$(2,1,1)$, 4	$(3,1)^2$, $(1,2)$	$-2, -2, 1$	$(1,2,2)$

In the first two rows $|\det X_k| = 1$, so the Smith form is $(1,1)$. In the third, X_4 is diagonalizable with both eigenvalues 0, so $X_4 = 0$. In the fourth and fifth, $|\det X_3| = 2$ is squarefree, which forces $(1,1,2)$ (this also follows from Corollary 2). In the last two, $|\det X_4| = 4$, and $X_4 \mp 2$ has rational rank 1, hence rank at most 1 over $\mathbb{F}_2$; as $X_4 \equiv X_4 \mp 2 \pmod 2$, at least two invariant factors are even, and the product 4 forces $(1,2,2)$. Every case agrees with the prediction. $\square$

6 Other readings

Other lattices. For $k = 3$, Theorem 1 covers every lattice stable under the permutations of $\{1, 2, 3\}$. Over the whole census the following were also compared. The dual lattice $\mathrm{Hom}(S^\lambda_\mathbb{Z}, \mathbb{Z})$, with the contragredient action, carries the transposed matrix, since transpositions are their own inverses; its Smith form is the same. The saturated lattice $S^\lambda_\mathbb{Q} \cap M^\lambda_\mathbb{Z}$, which contains $S^\lambda_\mathbb{Z}$, gives the same Smith form as $S^\lambda_\mathbb{Z}$ in all 135 cases with $n \leq 6$; it was not computed for larger n. The census also finds that λ and its conjugate λ' have the same Smith form for every k, in all 416 cases, as one expects from the classical relation between $S^{\lambda'}$ and the dual of S^λ twisted by the sign character [3, §8], under which X_k becomes $-X_k$.

The literal group ring reading. Read literally, the survey compares the Smith form of left multiplication by X_k on $\mathbb{Z}\mathfrak{S}_n$ with that of a block diagonal matrix holding f^λ copies of each predicted list. Grinberg's comment explains why this is not the intended statement, and it fails already for $n = 4$, $k = 4$. Writing d^m for m invariant factors equal to d, left multiplication by X_4 on $\mathbb{Z}\mathfrak{S}_4$ has Smith form $0^4\,1^{16}\,2^4$. The predicted lists are (3) for (4) and (1^4), $(1, 2, 2)$ for $(3, 1)$ and $(2, 1, 1)$, and $(0, 0)$ for $(2, 2)$; with multiplicities f^λ the diagonal holds $0^4\,1^6\,2^{12}\,3^2$, whose Smith form is $0^4\,1^8\,2^{10}\,6^2$. For $n \leq 3$ the two agree.

7 Corrected readings

We tested natural modifications of the formula against the same data, the Smith forms of X_k on $S^\lambda_\mathbb{Z}$ for all $\lambda \vdash n \leq 8$. Write N_c for the number of standard tableaux of shape λ in which k occupies the cell c, so that the formula as stated uses $S_r = \{c : N_c \geq r\}$. The readings are:

- *as stated*: $\alpha_{f-r+1} = |\prod_{c \in S_r} \mathrm{content}(c)|$;
- *multiset*: the Smith form of the diagonal matrix of these f products, ignoring their order;
- α_r *indexing*: the r-th product assigned to α_r instead of α_{f-r+1};
- *exactly r, more than r, at most r*: S_r replaced by $\{c : N_c = r\}$, $\{c : N_c > r\}$ or $\{c : N_c \leq r\}$;
- *multiplicity classes*: one product for each class of cells with equal N_c, assigned in increasing or decreasing order of N_c, or Smith-normalized;
- *eigenvalue diagonal*: the Smith form of the diagonal matrix of all eigenvalues of X_k on $S^\lambda_\mathbb{Q}$;
- *zero contents dropped*: the formula as stated, with cells of content 0 left out of the products;
- $X_{k\pm1}$: the formula for k compared with the Smith form of X_{k+1} or of X_{k-1};
- Y_k: the formula for k compared with the Smith form of $Y_k = \sum_{i>k}(i, k)$;
- λ': the formula for the conjugate shape λ' compared with X_k on S^λ;
- *fillings of* $1, \ldots, k$: N_c replaced by the number of standard fillings of $1, \ldots, k$ inside λ with k in c;
- *weak form*: every nonzero invariant factor is a product of some of the eigenvalues, the remark following the conjecture.

A case matches when the predicted list equals the Smith form entry by entry, taken nonnegative with zeros last. Where a reading has an ordered and a Smith-normalized version, the table gives the better count. The columns give the matches among the 240 cases with $n \leq 7$ and the 416 with

$n \le 8$ (196 and 350 for $X_{k\pm1}$, where $k \pm 1$ must lie in $1, \dots, n$), whether the reading reproduces Stanley's example $(1, 1, 3, 3, 12)$ for $\lambda = (5, 1)$, $k = 5$, and the first case, in order of n, where it fails.

reading	$n \le 7$	$n \le 8$	$(5,1)$, $k=5$	first failure
as stated	210	322	yes	$(3,1,1)$, $k=3$
multiset	210	322	yes	$(3,1,1)$, $k=3$
λ'	210	322	yes	$(3,1,1)$, $k=3$
α_r indexing	122	181	no	$(3,1)$, $k=3$
exactly r	96	133	no	$(2,1)$, $k=1$
more than r	47	70	no	(1), $k=1$
at most r	94	131	no	$(2,1)$, $k=1$
multiplicity classes	96	133	no	$(2,1)$, $k=1$
eigenvalue diagonal	177	279	yes	$(3,2)$, $k=4$
zero contents dropped	173	266	yes	$(3,2)$, $k=4$
X_{k+1}	3	3	no	(2), $k=1$
X_{k-1}	3	3	no	(2), $k=2$
Y_k	26	26	no	(2), $k=1$
fillings of $1, \dots, k$	144	203	yes	$(2,1)$, $k=1$
weak form	238	410	yes	$(5,2)$, $k=7$

The multiset reading agrees with the formula as stated in every case, since $S_1 \supseteq S_2 \supseteq \cdots$ makes the stated products a divisibility chain. The λ' reading agrees with it as well: conjugation moves the counts to transposed cells and changes the sign of the contents, and the census finds the same Smith form for λ and λ'. So does the reading with the labels counted from n downward, $Y_{n+1-k} = w_0 X_k w_0$ with w_0 the reversal permutation, whose Smith form equals that of X_k in all 416 cases.

No tested reading fits all the data with $n \le 7$. The best fit is the formula as stated, together with its equivalent readings; it matches 210 of the 240 cases with $n \le 7$ and first fails at $\lambda = (3, 1, 1)$, $k = 3$. Every other reading that predicts the whole Smith form first fails at $n \le 5$, most of them at $n \le 3$. The weak form, which predicts only a property of it, holds for all $n \le 6$ and fails at $n = 7$, for $(5, 2)$ and $(2, 2, 1, 1, 1)$ with $k = 7$ (Section 8). We propose no corrected form. The program for this comparison is included with the optional census programs; it needs the python-flint library.

8 The census for $n \le 8$

We computed the Smith form of X_k on $S^\lambda_{\mathbb{Z}}$ for every $\lambda \vdash n \le 8$ and every $1 \le k \le n$, 416 cases in all, and compared it with Conjecture 7.4. The numbers of disagreements are:

n	1	2	3	4	5	6	7	8	total
cases	1	4	9	20	35	66	105	176	416
disagreements	0	0	0	0	2	5	23	64	94

The two disagreements at $n = 5$ are those of Proposition 5. Stanley's example $\lambda = (5, 1)$, $k = 5$, is reproduced: the Smith form is $(1, 1, 3, 3, 12)$, as the survey states. Of the 94 disagreements, 23 have $k = 3$ and are accounted for by Corollary 2. Disagreements involve the prime 2 in 79 cases and the prime 3 in 30; they occur both for $k < n$ (85 cases) and for $k = n$ (9 cases), and for two-row shapes from $n = 7$ on, the first being $\lambda = (5, 2)$.

The weaker remark following the conjecture, that every SNF entry is a product of eigenvalues, also fails. For $\lambda = (5, 2)$ and $k = 7$ the entry 7 occupies $(1, 5)$ or $(2, 2)$, so the eigenvalues are 4 and 0, yet the Smith form is $(1, 1, 1, 1, 2, 4, 4, 4, 4, 0, 0, 0, 0, 0)$, against the prediction

$(1,1,1,1,1,4,4,4,4,0,0,0,0,0)$; the entry 2 is not a product of eigenvalues. The conjugate shape $(2,2,1,1,1)$ behaves in the same way. These are the first failures of the remark. Nor is the Smith form of X_k always that of the diagonal matrix of its eigenvalues: this holds in 279 of the 416 cases.

What survives in the data is limited: the conjecture holds for all $n \leq 4$ (Proposition 5) and at $k = 3$ for the shapes of Corollary 2, and the Smith forms of λ and λ' agree in every case. None of the corrected readings of Section 7 fits the data, and we propose none.

9 An open point

The documented facts are these. The survey reports that Grinberg computed the SNF of the matrices $W_{\lambda,k}$ for $\lambda \vdash n \leq 7$, and that the conjecture was made on the basis of these data; it defines $W_{\lambda,k}$ through a basis of $\mathbb{Z}\mathfrak{S}_n$. Grinberg's later comment says that it is defined "somewhat clumsily" and names the integral Specht module with its standard basis as the intended lattice. On that lattice, and by Theorem 1 on every lattice stable under the permutations of $\{1,2,3\}$, the conjecture fails at $n = 5$, and the census finds 30 disagreements with $n \leq 7$. The data of the computations reported in the survey are not available to us, and why they supported the conjecture is not known to us.

10 Verification

The interface. Three kinds of evidence are combined. Lean proves statements about explicit algebraic objects: the identity of Lemma 3, and, for the explicit integer matrix A, its Smith form, $AB = 2I$, and the unreachability of the predicted form. Standalone checkers establish that A is the matrix of X_3 on $S^{(3,1,1)}_{\mathbb{Z}}$ in the standard polytabloid basis, by recomputing it from the definitions. The general statements, Theorem 1, Corollary 2 and the proof of Proposition 5 for $n \leq 4$, are written proofs.

Certificates and checkers. The witness of Section 4, and a second witness $\lambda = (5,2)$, $k = 7$, are given as certificates, each holding the standard tableaux in order, the matrix A of X_k, unimodular U and V with UAV diagonal, the Smith form and the prediction. Two standalone checkers, written independently and using only the Python standard library, read them; neither trusts the encoded matrix. The first enumerates the standard tableaux from all $n!$ fillings, proves that the standard polytabloids form a $\mathbb{Z}$-basis of the $\mathbb{Z}$-span of all $n!$ polytabloids, recomputes A exactly from the tabloid module, checks $UAV = \mathrm{diag}(\mathrm{snf})$ with integer U, V of determinant ± 1 and a divisibility chain, and recomputes the prediction. The second, with separate code, enumerates the standard tableaux by lattice words, shows that the $\mathbb{F}_2$-span of all polytabloids has dimension f^λ, computes the rank of X_k there without reference to a basis, compares it with the number of odd predicted factors, verifies $X_k e_{T_j} = \sum_i A_{ij} e_{T_i}$ in the tabloid module for every j, and checks U, V and the Smith form with its own arithmetic. Both reject seven corrupted certificates (a changed entry of A; the Smith form replaced by the prediction; a non-unimodular U; a rational U of determinant ± 1; an extra factor; a non-standard basis tableau; an altered prediction), and both reject the agreeing case $\lambda = (5,1)$, $k = 5$, as showing no discrepancy. A third checker, written separately from both and importing neither, recomputes every $\lambda \vdash n \leq 5$ and every k: it proves that the standard polytabloids form a $\mathbb{Z}$-basis, forms the matrix of X_k, computes its Smith form from determinantal divisors, and asserts Proposition 5 and Stanley's example $\lambda = (5,1)$, $k = 5$. The entry point is `checks/check.py`, with the certificates in `data/`.

Census. The census was computed by two constructions. The first generates $S^\lambda_\mathbb{Z}$ inside $M^\lambda_\mathbb{Z}$ as the $\mathbb{Z}\mathfrak{S}_n$-span of one polytabloid, closing under simple transpositions with Hermite normal forms; it uses neither the standard basis nor straightening, and every Smith form is computed by two implementations. The second uses the standard polytabloid basis and exact rational linear algebra. They agree in all 135 cases with $n \le 6$. The census output is included as `data/census-n8.json`; the census programs are optional and not part of the check.

Lean. The checked statements are in Lean 4 [2] with mathlib [5], in the namespace `StanleySmithForms`: `group_ring_identity` states the identity of Lemma 3 in the integral group ring of $\mathfrak{S}_n$, for every $n \ge 3$; `A_det` states $\det A = -4$; `smith_form` states $UAV = \mathrm{diag}(1,1,1,1,2,2)$, and `U_det` and `V_det` give $\det U = -1$ and $\det V = 1$; `A_mul_B` and `B_mul_A` state $AB = BA = 2I$ for the explicit integer matrix $B = 2A^{-1}$; `rank_mod_two` states that A has rank 4 over $\mathbb{F}_2$; `not_predicted` states that no integer matrices P, Q with unit determinant satisfy $PAQ = \mathrm{diag}(1,1,1,1,1,4)$; and `counterexample` combines the Smith form and the unreachability of the predicted form. The passage from the identity, or from $AB = 2I$, to the invariant factors is the written argument of Section 3 and is not formalized. Lean does not check that A is the matrix of X_3 on $S^{(3,1,1)}_\mathbb{Z}$, that is, the construction of the Specht module and its standard basis, nor that $(1,1,1,1,1,4)$ is the predicted list; the certificate checkers, which recompute both from λ and k, and the derivation of Section 4 cover that step.

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